

Lecture 36: The Regulatory Path to Approval

The Oklo Aurora Case Study

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Introduction: The Traditional Path to Approval

For over 50 years, getting a new commercial nuclear reactor approved in the United States meant navigating the **Nuclear Regulatory Commission (NRC)**. This framework, codified in 10 CFR 50 and 52, relies heavily on prescriptive, deterministic safety analysis and one non-negotiable feature: a robust mechanism for containment of radioactivity, typically a reinforced concrete **Containment Dome**.

Under the NRC "gold standard," you assume a maximum credible accident (like a massive pipe break) and rely on a reinforced concrete structure to trap the radiation. This is highly effective for safety, but the approval process is notoriously cumbersome, expensive, and rigid. For emerging Small Modular Reactor (SMR) startups, the traditional NRC pathway presents a massive financial and bureaucratic wall.

In this lecture, we will examine how one company—Oklo—attempted to navigate this wall, failed, and ultimately pivoted to a completely new regulatory pathway, fundamentally altering their technology in the process.

1 Act I: The Original Aurora and the NRC Rejection

When Oklo first submitted their Combined License Application to the NRC in 2020, the "Aurora" was pitched as a revolutionary, highly innovative piece of technology.

1.1 The Innovative Design (Aurora v1)

The original Aurora design completely abandoned traditional light-water mechanics.

- **Micro-Scale Output:** The original Aurora was a true "microreactor," designed to produce just 4 MW of thermal power (MWt), yielding about 1.5 MW of electricity (MWe). To put this in perspective, the NRC application noted that the entire active core of the Aurora contained roughly the same volume of fuel as a *single fuel assembly* at a large commercial plant like Diablo Canyon.
- **The Fuel:** U-10Zr (metallic uranium-zirconium alloy) using High-Assay Low-Enriched Uranium (HALEU).
- **Heat Removal (Potassium Heat Pipes):** Instead of active coolant pumps or a massive pool of liquid metal, Oklo proposed a "forest" of sealed heat pipes. Each pipe contained a small amount of potassium working fluid operating passively at sub-atmospheric pressure.

As thermal superconductors, the potassium would vaporize at the bottom, travel upward, condense, and return via capillary action, offering redundant and reliable cooling for the tiny core.

- **Power Conversion (sCO₂):** To completely avoid the historic and highly explosive hazards of alkali metal-water reactions (such as potassium or sodium mixing with water), the heat pipes were coupled to a supercritical CO₂ (sCO₂) Brayton cycle rather than a steam turbine.
- **The Catch:** Oklo argued this system was so small, and inherently safe ("walk-away safe"), that they **did not need a containment dome**, proposing that their fuel provided **functional containment** and that they could house their reactor in a standard industrial weather structure (an A-frame building).

1.2 The Economic Motivation (The Square-Cube Law)

Why fight the NRC over the dome? Because of the Square-Cube law. For a large 1,100 MW reactor, a \$500 million concrete dome adds roughly \$5/MWh to the cost of electricity. But if you scale that down to a tiny 1.5 MWe to 50 MWe SMR, the power output (volume) shrinks much faster than the required surface area of the dome.

$$C \propto P^{2/3}, \quad \frac{C}{P} \propto P^{-1/3} \quad (1)$$

Scaling from 1,100 MW down to the SMR scale yields a massive unit-cost increase. A containment dome on a micro-reactor could add over \$15/MWh, rendering the plant completely economically uncompetitive against natural gas. Eliminating the dome was an economic mandate for Oklo.

1.3 The 2022 NRC Rejection

Oklo attempted to push this dome-less, micro-scale heat-pipe design through the rigid NRC. In 2022, the NRC denied the application.

- **The Reason:** The NRC demanded validated data and bounding analyses on what happens to metallic fuel during degraded heat-removal conditions, and what the quantitative source-term release would be if the cladding failed without a containment dome to catch it. Because Oklo relied on theoretical "functional containment" rather than empirical data, and because their application included gaps in safety analysis, design detail, and accident modeling, the NRC refused to proceed.

2 Act II: The Regulatory Pivot and the 2026 DOE Rewrite

Following the NRC rejection, the regulatory landscape in the US shifted dramatically. Seeking to bypass the NRC bottleneck, startups looked toward the Department of Energy (DOE), which has the authority to authorize experimental reactors on federal sites (like Idaho National Laboratory) outside of standard NRC commercial licensing.

2.1 The July 2026 Deadline and the NPR Analysis

In mid-2025, a presidential executive order mandated that the DOE approve at least three new experimental reactors to achieve criticality by July 4, 2026—an impossibly tight timeline in the nuclear world.

To make this happen, the DOE fundamentally rewrote its safety rules. As analyzed by NPR and ProPublica in early 2026, this rewrite represents a massive shift in regulatory philosophy:

1. **From "Shall" to "May":** The DOE systematically scrubbed prescriptive, mandatory language from its safety orders. Phrases dictating that operators "shall" ensure certain safety margins or that systems "must" survive specific transients were replaced with permissive wording ("may" or "should").
2. **Categorical Exclusions (CE):** The DOE granted advanced nuclear projects broad exemptions from standard National Environmental Policy Act (NEPA) environmental reviews.
3. **The Silicon Valley Ethos:** NPR reporting highlighted a deliberate push to inject a "move fast and break things" ethos into nuclear regulation, forcing out career NRC experts and relying heavily on the industry's own internal safety assessments rather than independent federal verification.
4. **Change in NRC Licensing:** While DOE can license "experimental reactors", the NRC is still responsible for commercial reactor licensing. In an NRC/DOE memorandum of understanding (August 2025) the NRC is now required to provide expedited commercial licensing to a reactor based on empirical results from DOE licensed prototypes. This circumvents the usual "what if" aspect of NRC evaluation.

3 Act III: The Design Pivot (Aurora v2)

Faced with the aggressive July 2026 DOE deadline, Oklo realized that getting their novel, unproven potassium heat-pipe and sCO₂ technology built and authorized in a matter of months was impossible. They needed historically proven technology.

3.1 Pivoting to an EBR-II Derived Design

The "new" Aurora is no longer a novel technology. It is fundamentally based on the Experimental Breeder Reactor II (EBR-II) that operated in Idaho for 30 years, with a significantly scaled-up power output.

- **A Massive Scale-Up:** The new design abandons the tiny 4 MWt microreactor concept. The revised Aurora targets 15 MWe to 50 MWe (roughly 50 to 150 MWt). This represents a 10× to 30× increase in power output and physical size, fundamentally changing the decay heat and source-term calculations.
- **Pool-Type Sodium:** They abandoned the potassium heat pipes for a standard liquid sodium pool and intermediate sodium loop architecture.
- **Steam Turbine:** Because commercial sCO₂ turbines do not readily exist, Oklo signed a contract with Siemens Energy to use a standard off-the-shelf SST-600 steam turbine.

3.2 The Branding Question: Why Keep the Name?

If the reactor is 30× larger, uses a massive pool of liquid sodium instead of potassium heat pipes, and drives a steam turbine instead of a supercritical CO₂ cycle—why keep the name “Aurora”?

A likely explanation lies not in reactor physics, but in the practical realities of project development and finance. Maintaining continuity in project identity can help avoid complications such as renegotiation of power purchase agreements, re-evaluation of early-stage site permits, and potential uncertainty among investors who committed capital based on the original project concept and branding.

In this sense, the persistence of the “Aurora” name reflects the extent to which non-technical constraints—including schedule, financing, and regulatory positioning—can shape both the design and presentation of engineering systems.

4 Act IV: The Remaining Technical Risk

Oklo successfully pivoted their regulatory approach (from the strict NRC to the permissive DOE) and their design approach. However, they **still refuse to build a containment dome**, relying on the relaxed DOE rules to allow a standard industrial building. This introduces a major, historically documented risk:

4.1 Direct Sodium-Water Coupling Without a Dome

By reverting to an EBR-II style pool and a standard steam turbine, Oklo brought back the oldest engineering hazard in fast reactors: the sodium-water reaction. If a tube leaks in the steam generator, high-pressure steam will mix with secondary sodium, producing explosive hydrogen gas, highly caustic sodium hydroxide, and massive pressure spikes:



While the original EBR-II utilized a steel containment dome to seal in the reactor (though not a reinforced concrete pressure dome), Oklo’s Aurora proposes a standard industrial weather structure. Under the new “permissive” DOE regulations, the success of the Aurora pilot will test whether an un-domed sodium plant can truly be operated safely, or if the classic NRC “gold standard” was right all along.

5 Act V: The Regulatory Shift: From Prescriptive to Discretionary

Following the 2025 executive actions aimed at accelerating experimental reactor deployments, the Department of Energy (DOE) revised portions of its safety and operational rules.

A notable feature of these revisions is a shift in language from prescriptive requirements (“shall,” “must”) toward more discretionary or performance-based phrasing (“may,” “as appropriate”). Such changes do not eliminate safety requirements, but they can alter how those requirements are implemented, verified, and enforced. In this section, we examine several specific rule changes as cited by NPR and analyze their potential engineering implications.

5.1 Case Study 1: The Groundwater Rules

Text of the Rule Change

CURRENT RULES (DOE O 458.1):

PROTECTION OF DRINKING WATER AND GROUND WATER

Ground water must be protected from radiological contamination to ensure compliance with dose limits in the Order and consistent with ALARA process requirements. To this end, DOE must ensure that: (a) Baseline conditions of the ground water quantity and quality are documented; (b) Possible sources of, and potential for, radiological contamination are identified and assessed; (c) Strategies to control radiological contamination are documented and implemented; (d) Monitoring methodologies are documented and implemented; and (e) Ground water monitoring activities are integrated with other environmental monitoring activities.

NEW RULES (NE O 458.1A):

GROUND WATER

Consideration must be given to avoiding or minimizing potential radiological contamination of groundwater from NE-authorized nuclear facilities and radiological activities, to ensure compliance with public dose limits in this Order. To this end: (i) baseline conditions of the ground water quantity and quality may be considered; (ii) potential sources of radiological contamination may be identified; (iii) strategies to control radiological groundwater contamination may be developed, as appropriate; (iv) monitoring may be implemented, as appropriate, and integrated with other monitoring activities.

Engineering Implication

The revised language removes explicit prescriptive requirements for baseline characterization, monitoring, and documentation, replacing them with discretionary phrasing such as “may be considered” and “as appropriate.” It also eliminated any reference to the ALARA principle “As Low as Reasonably Achievable,” a rather startling change.

From an engineering and environmental monitoring perspective, baseline data play a critical role in attribution. Without well-characterized pre-operational conditions, it becomes significantly more difficult to distinguish facility-induced contamination from pre-existing or background conditions using standard methods such as trend analysis or plume reconstruction.

Similarly, discretionary monitoring requirements may lead to increased variability in how different facilities implement environmental surveillance programs. While monitoring is not necessarily eliminated, the shift in language transfers greater responsibility to site-specific judgment and regulatory interpretation.

5.2 Case Study 2: Accident Investigations and Dose Thresholds

Text of the Rule Change

CURRENT RULES (DOE O 225.1B):

LOSS OF CONTROL OF RADIOACTIVE MATERIAL

(1) Any single accident that results in: (a) A general employee exceeding any of the external dose limits in 10 C.F.R. Part 835.202, Occupational Dose Limits for General Employees, by a factor of two or more.

NEW RULES (NE O 225.1):*LOSS OF CONTROL OF RADIOACTIVE MATERIAL*

(1) Any single accident that results in: a. A general employee exceeding any of the external dose limits in 10 C.F.R. Part 835.202, Occupational Dose Limits for General Employees, by a factor of four or more.

Health Physics Implication

The standard occupational dose limit for radiation workers is 5 rem/year. Under the previous rule, a single-event exposure exceeding 10 rem triggered formal investigation; the revised rule raises this threshold to 20 rem.

From a biological standpoint, an acute dose of 20 rem (0.2 Sv) is generally below the threshold for acute radiation syndrome (ARS), which typically begins near 100 rem. However, such a dose is well above normal operational expectations and represents a significant radiological control event.

From an engineering perspective, exposures of this magnitude typically indicate a breakdown in one or more layers of radiological protection, such as shielding, procedural controls, or access restrictions. The lower investigation threshold historically served as an early-warning trigger to identify and correct such failures before they could escalate.

Raising the threshold does not change dose limits themselves, but it does alter when formal investigation is required. This may delay systematic root-cause analysis in certain scenarios, depending on how the revised criteria are implemented.

5.3 Case Study 3: Physical Security and Protective Forces**Text of the Rule Change****CURRENT RULES (DOE O 473.2A):***PERSONAL PROTECTIVE ARMOR*

Personal protective armor must be issued for all SPOs [Security Police Officers]. (1) Protective armor for SPOs must provide Type III protection, at a minimum... (2) Protective armor must be worn at all times by: (a) SPOs assigned posts that interact with the general public... (b) SPOs at sites with PL 1-4 assets. The ODFSA may waive this requirement for specific posts... This waiver must be documented.

NEW RULES (NE O 470.1):*(A) PERSONAL PROTECTIVE ARMOR.*

The contractor will: i. Issue personal protective armor for all SPOs as identified or required through local analysis. ii. Ensure issued protective armor for SPOs provides Type III protection, at a minimum... iii. Wear protective armor as directed by the ODSA.

Operational Implication

The revised rule shifts decision-making authority regarding protective posture from a centralized requirement to site-specific analysis. Rather than mandating universal use of protective armor in defined situations, the new framework allows deployment decisions to be based on local threat assessments.

From a security engineering perspective, nuclear facilities are traditionally designed using a *defense-in-depth* approach, incorporating multiple layers of protection including physical barriers,

armed response, and deterrence measures. Standardization across sites can contribute to predictable baseline security levels. Simply put, it helps make a nuclear facility a “hard target,” contributing to deterrence by increasing the perceived difficulty and risk of adversary actions.

Introducing site-specific discretion may allow operators to tailor security posture to local conditions, but it may also lead to variability in implementation across facilities. The engineering question becomes one of consistency: whether decentralized decision-making maintains equivalent levels of protection across a distributed fleet of reactors.

5.4 Case Study 4: Protection of Biota and Ecological Impact

Text of the Rule Change

CURRENT RULES (DOE O 458.1):

PROTECTION OF BIOTA.

(1) Radiological activities that have the potential to impact the environment must be conducted in a manner that protects populations of aquatic animals, terrestrial plants, and terrestrial animals in local ecosystems from adverse effects due to radiation and radioactive material released from DOE operations.

(2) When actions taken to protect humans from radiation and radioactive materials are not adequate to protect biota then evaluations must be done to demonstrate compliance with paragraph 4.j.(1) of this Order in one or more of the following ways: (a) Use DOE-STD-1153-2002, A Graded Approach for Evaluating Radiation Doses to Aquatic and Terrestrial Biota. (b) Use an alternative approach to demonstrate that the dose rates to representative biota populations do not exceed the dose rate criteria in DOE-STD-1153-2002, Table 2.2. (c) Use an ecological risk assessment to demonstrate that radiation and radioactive material released from DOE operations will not adversely affect populations within the ecosystem.

NEW RULES (NE O 458.1):

BIOTA

Consideration may be given to avoiding or minimizing, if practical, potential adverse impacts to aquatic animals, terrestrial plants, and terrestrial animals in local ecosystems from radiation and releases of radioactive material, using a graded approach.

Environmental Engineering Implication

The revised language removes the explicit requirement for a standardized, quantitative evaluation framework for protection of non-human biota. Under the previous rule, if baseline protections for human health were not sufficient to protect the surrounding ecosystem, facility designers were required to demonstrate compliance using defined methods, such as the dose rate criteria in DOE-STD-1153-2002 or a formal ecological risk assessment.

In contrast, the new rule replaces these prescriptive requirements with discretionary language such as “consideration may be given” and “if practical,” without reference to specific dose limits or required analytical methods. This represents a shift from a compliance-based framework toward a more performance-based or judgment-based approach.

From an engineering perspective, the introduction of “if practical” implicitly incorporates feasibility, cost, and project constraints into decisions about environmental protection. As a result, the extent to which ecological impacts are quantitatively evaluated or mitigated may vary across facilities, depending on site-specific decisions and regulatory interpretation.

The removal of explicit dose criteria as a required design check may also reduce the likelihood that protection of non-human biota serves as a primary driver for the inclusion of dedicated monitoring or mitigation systems. While environmental protection is not eliminated, it may become less uniformly defined and less explicitly quantified than under the previous framework.

More broadly, this change illustrates a recurring theme in the revised regulatory structure: a transition from prescriptive, standardized requirements toward flexible, case-dependent implementation. The engineering implications of this shift depend not only on the written rules, but on how rigorously they are applied in practice.

5.5 Summary

Across these examples, the common feature is not the removal of safety requirements *per se*, but a shift in how those requirements are specified. Prescriptive rules provide uniformity and enforceability, while discretionary rules allow flexibility and adaptation.

From an engineering standpoint, this shift transfers greater responsibility to:

- Site operators, who must interpret and implement requirements
- Regulators, who must evaluate compliance under less rigid criteria
- Designers, who must account for a wider range of acceptable implementations

The long-term impact of this transition will depend not only on the written rules, but on how they are applied in practice. In particular, greater flexibility may lead to increased variability in how safety margins and environmental protections are implemented across different projects.

“If you lived downwind of the Aurora Powerhouse, would you prefer a visually appealing glass-walled reactor building with no large containment structure, or a reinforced containment dome designed for worst-case accidents—and why?”

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